

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Project Type

Group / Individual Project

Subject

Cybersecurity Law & Policy

DeliverablesReport + Presentation + Flowcharts +
Infographics**Total Marks**

100 Marks

Important Notice to Students

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

Name:	Areeba Ashfaque Ahmed Khan
Roll Number	25102C2009
Email id	Khanareeba3007@gmail.com
Contact Number	8097828676

Name:	Juveriya Mohd. Shuaib Khan
Roll Number	25101C2002
Email id	juveriya363@gmail.com
Contact Number	9137510678

PART 1: Introduction

Overview

VIT's has over 50 million internet users. VIT's has online banking, ecommerce, cloud based businesses, smart city infrastructure, and AI public services. Although these technologies provide conveniences and stimulate the economy, they also provide increasing security and cyber concerns for the country.

VIT's has experienced many different kinds of cyber threats. There have been breaches, ransomware attacks on hospitals, financial cyber crimes, identity theft, and cyber bullying. There have been attacks on social media and other platforms with false information and manipulation. This shows that there is a need for a comprehensive and strong legal framework for the protection of citizens and the infrastructure of the country.

Purpose of Law

The framework of the National Cybersecurity Law is for the creation of a safe digital environment. The personal data of citizens would be protected, cybercrime would be prevented, and important infrastructure would be protected. The Law would also provide a framework for the promotion of safe and responsible technology. The Law provides framework for enforcement and compliance of rights, enhanced responsibilities, penalties, and increased cybersecurity.

Importance of Cybersecurity

VIT's increasing Reliance on Digital Services makes Cybersecurity a primary focus. Cybersecurity is of value to National Security, Economic Stability and Public Trust. Cybersecurity is the protection of critical services, loss of revenue, and the protection of sensitive data. Safe Digital Development mandates the presence of Cybersecurity Law.

Challenges Faced by VIT's

- Cyber concerns for VIT's are many. They include:
- Data breaches of citizens' personal information
- Ransomware attacks on hospitals
- Banking cyber fraud
- Identity theft and hacked accounts
- Misinformation on social media
- Cyberbullying and harassment

These threats are to every citizen, every enterprise, and every government institution.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To Be Completed by Student)
1	Cybercrime	<i>Cybercrime is any criminal activity that targets, or uses a computer, a computer network, or a networked device to commit an offense. While some crimes aim to damage systems, the vast majority are carried out by individuals or organized groups for financial gain.</i>
2	Hacker	<i>Hacker is a person who uses a computer to look at and/or change information on another computer without permission</i>
3	Ethical Hacker	<i>An ethical hacker is a cybersecurity professional authorized to legally hack into computer systems and networks. Also known as a "white hat" hacker, their primary job is to think and act like a malicious attacker to identify security vulnerabilities before cybercriminals can exploit them.</i>
4	Malware	<i>Malware, or "malicious software," is an umbrella term for any program or code intentionally designed to damage, disrupt, or gain unauthorized access to a computer system. Cybercriminals use it to steal sensitive data, extort money, or hijack device operations.</i>
5	Ransomware	<i>Ransomware is a type of malicious software (malware) that encrypts a victim's files or locks their device, rendering them unusable. Attackers then demand a financial payment in exchange for the decryption key or unlock code, threatening to permanently delete or publicly leak the data if the ransom is not paid.</i>
6	Phishing	<i>Phishing is a form of social engineering where attackers impersonate trusted entities—such as banks, employers, or popular brands—to trick individuals into revealing sensitive information, such as passwords or credit card details, or to install malware.</i>
7	Data Breach	<i>A data breach is a security incident in which unauthorized individuals access, steal, or leak sensitive, confidential, or protected data. It can happen to individuals or organizations, and involves exposing information like personal identities (PII), financial records, or corporate intellectual property to untrusted environments.</i>
8	Personal Data	<i>Personal data, or personal information, is any information that can be used to identify a living individual, either directly or indirectly.</i>
9	Sensitive Data	<i>Sensitive data is any information that requires protection against unauthorized access, disclosure, or misuse because its exposure could cause harm, discrimination, or financial loss.</i>
10	Digital Signature	<i>A digital signature is a cryptographic mathematical scheme used to verify the authenticity, integrity, and non-repudiation of a digital message or document.</i>
11	Electronic Evidence	<i>Electronic evidence (or digital evidence) is any information of probative value stored or transmitted in binary/digital format that can be used in legal or investigative proceedings. This includes data like emails, text messages, browsing history, audio/video files, and GPS logs. [1, 2, 3]</i>

1 2	Identity Theft	<i>Identity theft occurs when someone wrongfully obtains and uses your personal information—such as your name, ID numbers, or financial details—without your permission to commit fraud or other crimes.</i>
1 3	Artificial Intelligence	<i>Artificial intelligence (AI) is the simulation of human intelligence by computers and machines. It enables systems to learn from data, reason, solve problems, and perform cognitive tasks that typically require human intellect, such as understanding language and recognizing patterns.</i>
1 4	Cloud Computing	<i>Cloud computing is the on-demand delivery of computing services—including servers, storage, databases, networking, and software—over the internet.</i>
1 5	Critical Infrastructure	<i>Critical infrastructure refers to the physical and virtual systems, assets, and networks that are so essential to a society that their incapacitation or destruction would have a debilitating impact on national security, the economy, public health, or safety.</i>
1 6	Cyberbullying	<i>Cyberbullying is the use of digital technologies—like social media, messaging apps, and online gaming—to repeatedly harass, threaten, embarrass, or target another person.</i>
1 7	Deepfake	<i>A deepfake is synthetic media (video, audio, or images) generated or heavily manipulated by artificial intelligence. The name combines "deep learning" (a type of AI) and "fake."</i>
1 8	Encryption	<i>Encryption is the process of scrambling readable data into unreadable code to prevent unauthorized access. It acts as a digital lock, converting "plaintext" into "ciphertext".</i>
1 9	Firewall	<i>A firewall is a network security device or software that monitors and filters incoming and outgoing traffic.</i>
2 0	Zero-Day Exploit	<i>A zero-day exploit is a cyberattack that targets a software or hardware vulnerability that is unknown to the vendor or security community. It is called "zero-day" because the developers have had zero days to create a patch or defense, giving attackers the advantage of surprise.</i>

PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Citizens must be protected from the unauthorized collection, use, or sharing of their personal data. Every citizen of VIT has the fundamental right to privacy in the digital environment. No government agency, private organization, or individual may collect, process, store, monitor, or share a citizen's personal data without a legal reason and informed consent, except when required by law for national security or criminal investigations. Personal data includes names, addresses, contact information, identification numbers, online activities, and biometric information.

Organizations must clearly inform citizens about the type of data being collected, the purpose of collection, how long it will be stored, and any third parties that may receive the data. Citizens must be able to withdraw consent at any time unless the law requires retention.

This right will be enforced by the National Cyber Security Authority (NCSA). Organizations that violate privacy rights may face fines of up to 5% of annual revenue and be required to take corrective actions. Citizens may file complaints with the NCSA, which must investigate and resolve cases within 60 days.

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures. Each individual is entitled to have the assurance that any organization collecting his or her data implements effective safeguards to ensure that it is not accessed, disclosed, altered, or destroyed without authorization. Each public and private entity operating within VIT's jurisdictions shall be required to deploy industry best practices in cybersecurity which include encryption, multifactor authentication, access control policies, and routine security audits.

Annual assessments and penetration tests for vulnerabilities should be conducted. The data such as financial data, healthcare records, and biometric data should be afforded increased protection using robust encryption methods and stringent access controls.

The NCSA will periodically conduct compliance and security audits. Entities found wanting in protecting the data of the citizens may incur penalties, suspension of business, and corrective measures. Where there is negligence and the resulting data breach affects the citizens, they are entitled to seek damages.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems. People are allowed to ask organizations to delete all their data once the information is no longer required for the initial purpose, people withdraw their consent, or the

data collection has been conducted illegally. The term that is often used to describe the aforementioned right is “Right to Be Forgotten.”

The organizations will need to reply within a period of 30 days, either deleting the information or offering valid reasons to keep it. Information that is deleted will be taken out of the backups and third parties as far as it is possible.

NCSA will be responsible for creating one point of contact for citizens who find themselves ignored by organizations. Penalties will be substantial if organizations do not comply with these regulations.

Right to Information

Citizens have the right to know what data is collected about them and how it is used. The citizens have the right to be aware of what personal data is being collected, how such information is used, where it is stored, and with whom it has been shared. The organizations should be obliged to offer understandable privacy notices prior to collecting any personal data.

On being requested by citizens, organizations will be obliged to offer copies of such data in a commonly used digital form without charging anything from the citizen for such purposes. In addition, the citizens should be notified about automated decision-making systems, which include artificial intelligence profiling systems.

The NCSA shall regulate compliance with such transparency regulations and handle complaints related to lack of transparency accordingly.

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes. All citizens have the right to report cases of cybercrime without the fear of any form of persecution, discrimination, coercion, or legal threats. Cases of cybercrime can be reported by citizens and include phishing, identity theft, cyberbullying, ransomware attacks, online fraud, and hacking of accounts.

The government shall establish a round-the-clock portal and hotline for reporting any case of cybercrimes. Within 24 hours, the complainant should receive an acknowledgment and within seven days receive updates regarding the preliminary investigations carried out.

The NCSA and all law enforcement agencies will be responsible for maintaining professionalism while handling all complaints. An official ignoring any legitimate cybercrime complaint is liable to disciplinary action.

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation. It shall be assumed that an individual who is suspected of conducting a cybercrime is innocent unless proven otherwise by a competent court of law. Citizens also have the right to receive fair and prompt investigations through proper channels within the limits of the law.

Law enforcers are only supposed to access personal gadgets, accounts, and messages when they have the appropriate authority from the law. This is because digital evidence can

only be collected and stored through legal forensic techniques.

The National Cybersecurity Authority shall supervise all cybercrime investigations to ensure justice. In cases where the citizens are falsely accused, they have the option of filing for damages in the Cybersecurity Law of VIT.

Right to Digital Consent

The right of the citizens to give informed and voluntary consent should exist prior to the collection and/or processing of their personal information.

Requests for consent should be made using clear language and not hidden in long sets of terms and conditions.

Consent choices should be offered separately by organizations depending on the type of processing of personal information done, and citizens should be allowed to withdraw from their consent anytime.

The NCSA will review the processes used for obtaining consents and penalize organizations found to be using deceitful means to solicit consent.

Right to Protection of Children Online

Children below 18 years old shall have increased safeguards in the digital space. Companies that offer online services to children should adopt adequate security and privacy controls and seek consent from the children's parents where necessary.

Online platforms that target children shall ban cyberbullying, grooming, exploitation, and other online harms. Educational institutions shall provide training programs on cybersecurity to enable their students to recognize online dangers.

The NCSA should collaborate with schools, parents, and technology companies in enforcing these child protection measures. Any violations relating to children shall attract stricter consequences and swift investigation.

PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Unauthorized access occurs when a person gains entry to a computer system, network, account, or database without permission. Example: A university student in VIT's hacks into the National Education Portal and accesses confidential student records.	3–5 years imprisonment	₹500,000
Data Theft	Data theft involves the illegal copying, downloading, transfer, or sale of confidential information. Example: An employee of a cloud company steals customer financial records and sells them on the dark web.	5–8 years imprisonment	₹1,500,000
Identity Theft	Identity theft occurs when someone uses another person's personal information to impersonate them for financial or criminal gain. Example: A criminal uses a citizen's digital identity card details to open bank accounts and obtain loans.	4–7 years imprisonment	₹1,000,000
Phishing	Phishing involves fraudulent emails, messages, or websites designed to trick victims into revealing sensitive information. Example: Citizens receive fake messages claiming to be from the National Bank of VIT's requesting login credentials.	3–6 years imprisonment	₹750,000
Ransomware	Ransomware is malicious software that encrypts data and demands payment for its release. Example: Attackers encrypt patient records at a major hospital and demand cryptocurrency before restoring access.	8–15 years imprisonment	₹5,000,000
Malware Distribution	Malware distribution refers to creating, spreading, or selling malicious software that	5–10 years imprisonment	₹2,000,000

	damages systems or steals information. Example: A cybercriminal distributes infected software updates that compromise thousands of computers across VIT's.		
Cyberbullying	Cyberbullying involves using digital platforms to repeatedly harass, threaten, humiliate, or intimidate another person. Example: Students repeatedly post abusive comments and edited images targeting a classmate on social media.	Up to 3 years imprisonment	₹300,000
Online Harassment	Online harassment includes threats, stalking, intimidation, or offensive communications through digital means. Example: An individual repeatedly sends threatening messages to a journalist investigating corruption.	2–5 years imprisonment	₹500,000
Financial Fraud	Financial fraud involves using digital systems to illegally obtain money or financial benefits. Example: Criminals create a fake investment application and steal funds from thousands of citizens.	7–12 years imprisonment	₹3,000,000
Misinformation	Misinformation involves intentionally spreading false information that causes public panic, economic disruption, or social unrest. Example: Fake posts claim that major banks in VIT's have collapsed, causing mass panic withdrawals.	2–5 years imprisonment	₹1,000,000
Website Defacement	Website defacement occurs when attackers alter a website's content without authorization. Example: Hackers replace the homepage of the Ministry of Health website with political propaganda.	4–8 years imprisonment	₹1,500,000
Cyber Terrorism	Cyber terrorism involves attacks against critical infrastructure intended to threaten national security, public safety, or government operations.	15–25 years imprisonment or life imprisonment in severe cases	₹10,000,000

	Example: A terrorist group launches cyberattacks against power grids, airports, and emergency services, causing nationwide disruption.		
--	---	--	--

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

Organizations should limit the data collected on individuals to that which is required for rendering a particular service or for carrying out a particular activity that is justified by business needs. It should be lawful and legitimate; it should also be for a well-defined purpose. An organization must inform individuals of the type of data that would be collected, the purpose of collecting the data, period of its retention, and parties to whom the data would be disclosed.

Sensitive information such as biometric data, financial data, medical data, and identity numbers issued by government authorities can be collected if necessary but only with the consent of the individual. Organizations should not collect any unnecessary or irrelevant data.

Every organization collecting personal data should keep record of its data collecting activities and should provide the same to NCSA on demand. Unlawful data collection can result in penalties up to ₹2,000,000 and stoppage of data processing activities.

Data Storage Rules

All sensitive information needs to be stored safely through cybersecurity control measures that are in line with industry standards. Organizations will put in place measures like encryption, firewalls, intrusion detection, access controls, and backups to ensure that the stored information is safe from any form of tampering.

Sensitive information will be encrypted when in transit and when it is stored. Access to stored information will only be allowed by people who have been authorized to access it depending on organizational needs. Audit logs of all access attempts will be maintained for at least two years.

Organizations will perform annual security assessments and vulnerability tests to ensure that the storage security controls are effective.

Data Sharing Rules

Organizations will only disclose the personal information of citizens to third parties if the process is absolutely necessary for service delivery, required by law, or permitted by informed consent of the user. All data-disclosure processes must involve legal contracts that provide similar standards of security and confidentiality as that of organizations.

Organizations must conduct security assessments before disclosing personal information to third parties to check if they have the capability of protecting the data. The data shared with third parties must be encrypted. If the contracted third party misuses the information of citizens, then it will still be the responsibility of the organization. Any disclosure of personal information without permission can lead to severe monetary and other punishments.

User Consent Requirements

It is necessary that an organization gets explicit and free consent of individuals before collecting, processing, storing, or transferring their personal information. A citizen should have the right to revoke the consent at any point of time without facing any kind of discrimination or denial of service. For the collection of personal information, parental or guardian's consent will be necessary if the individual is below the age of 18 years. There can be a fine of up to ₹1,500,000 in case there is any failure in obtaining proper consent.

Data Retention Period

Organizations shall retain personal data only for as long as necessary to fulfill the purpose for which it was collected. Once the purpose has been completed, the information must be securely deleted unless retention is required by law, contractual obligations, or ongoing investigations.

The following maximum retention periods shall apply:

- * Customer account information – 5 years after account closure
- * Financial transaction records – 7 years
- * Employee records – 5 years after employment termination
- * Medical records – 10 years
- * Security logs and audit records – 2 years
- * Marketing information – Until consent is withdrawn or after 3 years of inactivity

Organizations must maintain documented retention schedules and periodically review stored information to ensure compliance.

Data Deletion Process

The citizen can demand the deletion of his/her personal data in case the data is no longer needed, if there is withdrawal of consent, or if the data was acquired illegally. There should be a convenient procedure in place for making such deletion requests.

After receiving a proper request, the organization should verify the identity of the requester and delete the data within 30 days. The deleted data should be completely deleted from all active files and archives. Confirmation should be provided to the user after the deletion has been done. Ignoring the request of the user could lead to regulatory action against the organization, along with compensation claims.

Breach Notification Requirements

Any organization experiencing a data breach involving unauthorized access, disclosure, loss, theft, or destruction of personal information must immediately activate its incident response procedures.

The organization must notify the National Cyber Security Authority (NCSA) within 72 hours of discovering the breach. The notification shall include:

- * Nature of the breach
- * Number of affected individuals
- * Types of data compromised
- * Likely risks to citizens
- * Mitigation measures being implemented

Affected citizens must be informed without unreasonable delay whenever the breach is likely to result in identity theft, financial loss, privacy violations, or other significant harm. Organizations that intentionally conceal breaches may face fines up to ₹5,000,000, suspension of operations, and criminal liability for responsible executives.

Framework Enforcement

The National Cyber Security Authority (NCSA) shall monitor compliance with this Data Protection Framework through inspections, audits, investigations, and citizen complaints. Organizations found violating data protection requirements may face:

- * Financial penalties
- * Mandatory corrective actions
- * Suspension of data processing activities
- * Revocation of operating licenses
- * Civil and criminal liability for severe violations

This framework shall apply to all organizations operating within VIT's, regardless of size or sector, whenever they collect, process, store, or manage citizen data.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: Data Protection and Privacy Compliance

Personal data must always be gathered, processed, and stored for lawful and valid reasons and with the explicit consent of the citizens. The personal data must not be sold or made available to any third party that is not legally authorized to do so.

This provision applies to all organizations that handle citizen data. Non-compliance will lead to fines of up to ₹2,000,000 and also stoppage of processing activities.

Responsibility 2: Encryption of Sensitive Data

It is mandatory for any organization to encrypt all their confidential data such as financial records, medical records, biometric data, and government data while storing and transmitting such data. Encryption should take place using industry standard encryption protocols.

Such a provision will apply to all organizations dealing with such data. Any breach of this provision leading to data leakage can attract a fine of ₹3,000,000 or more.

Responsibility 3: Multi-Factor Authentication (MFA)

All organizations must put in place the Multi-Factor Authentication feature for all their administrative accounts, privileged users, and any system that contains confidential data.

This requirement is applicable to all mid-sized and large organizations. Non-compliance will lead to regulatory sanctions to the tune of ₹1,000,000.

Responsibility 4: Regular Security Audits

Companies are required to perform independent cybersecurity audits every year in order to identify any vulnerability and check if they comply with the national cybersecurity norms. The audit report needs to be sent to the NCSA on demand.

This is applicable to companies with more than 100 workers or which handle confidential information. There will be penalties of up to ₹1,500,000 for non-compliance along with additional compliance inspections.

Responsibility 5: Employee Cybersecurity Training

All organizations are required to offer mandatory cyber security awareness training to their employees at least twice per year. These trainings will cover phishing awareness, password protection, social engineering, and incident reporting protocols.

All employers are obliged to do this. Those who fail to train their employees will have to pay a penalty of up to ₹750,000.

Responsibility 6: Cyber Incident Reporting Within 72 Hours

Reporting of any security breach in terms of any sort of hacking, ransomware attacks, or data theft in VIT's premises should be done within 72 hours of occurrence to the National Cyber Security Authority.

Failure to comply with the said regulation may attract a fine of not less than ₹5,000,000. This requirement applies to all firms operating within VIT's jurisdiction.

Responsibility 7: Appointment of a Data Protection Officer (DPO)

Data Protection Officers have to be appointed by organizations handling more than 10,000 citizens' information. The DPO will be the main point of contact for regulatory and complaints.

Failure to appoint a Data Protection Officer can lead to a fine of up to ₹1,000,000.

Responsibility 8: Annual Penetration Testing

Penetration tests must be performed at least annually in organizations using cybersecurity experts who are duly certified. Testing is aimed at detecting weaknesses that could later be exploited by hackers.

This mandate is applicable to financial companies, healthcare companies, cloud computing services providers, and government contractors. Non-compliance will incur penalties of up to ₹2,000,000 and mandatory security tests.

Responsibility 9: Access Control and Least Privilege Policy

It is imperative for organizations to adopt role-based access control mechanisms that will guarantee that only information necessary for doing one's job is accessible to the employees.

The need to perform this obligation is binding on all organizations managing sensitive information. The failure to comply with this obligation due to excessive access to information can lead to penalties of up to ₹1,500,000.

Responsibility 10: Incident Response and Business Continuity Planning

The organizations are supposed to keep an IRP and BCP document. The plans must include procedures for identifying, reacting, restoring, and reporting cyber security attacks.

The organization must conduct annual simulation or disaster recovery exercise of their plans. The non-maintenance of appropriate plans leads to the imposition of penalty up to ₹2,500,000.

Responsibility 11: Secure Backup and Recovery Systems

The organizations are to ensure that there is proper backup of the vital information belonging to business and customers. There should be tests done to guarantee proper recovery after any cyber-attack like the ransomware attack.

Any organization not ensuring proper backup will be subject to a fine of up to ₹1,500,000.

PART 7: Government Responsibilities

Responsibility for protecting the state's security, its safety and economic stability, along with critical cyber-infrastructure, shall lie on the Government of VIT. All government organizations shall work with the National Cyber Security Authority (NCSA) to ensure that appropriate measures are put into place.

Protect National Infrastructure

The Government of VIT's will identify and classify key sectors within critical infrastructures whose failure will pose risks to the country's national security, economic stability, and public safety. Critical infrastructures include:

- * Power generation and power grid systems
- * Water supply and water treatment systems
- * Hospitals and health care infrastructures
- * Banks and financial infrastructures
- * Transportation infrastructures including airports and railways
- * Telecommunication networks
- * Government infrastructures and e-governance services
- * Emergency services

All critical infrastructure organizations are expected to meet stringent cybersecurity standards which will be put in place by the NCSA. The cybersecurity standards shall include network segmentation, data encryption, ongoing monitoring, vulnerability testing, backups, and disaster recovery services.

The government will undertake an annual audit of all critical infrastructure facilities for security purposes. Those organizations that will fail to meet the required security standards will face regulatory sanctions.

Monitor Cyber Threats

The Government of VIT's will develop a National Cyber Threat Monitoring Center that will operate 24 hours, 7 days a week.

The monitoring tools will include:

- * Security Information and Event Management (SIEM) tools
- * Threat Intelligence Platforms
- * Artificial Intelligence-based Threat Detection tools
- * Intrusion Detection and Prevention Systems (IDPS)
- * National cyber incident reporting systems

The NCSA will analyze and share cyber threat intelligence among appropriate stakeholders to preempt any attacks from occurring. The government departments will ensure that communication is done in real time regarding emerging cyber threats.

The monitoring process will not infringe on the privacy rights of the citizens and will only involve collection of information that is necessary for national cybersecurity as per the law.



Investigate Cybercrimes

The Government of VIT's shall set up dedicated Cybercrime Investigation Units with digital forensics experts, cybersecurity professionals, and law enforcement officers.

The investigation of cybercrimes shall be undertaken in accordance with legal procedures, which include:

Step 1: Compliant Registration

A citizen or organization can report cyber incidents via the National Cybercrime Reporting Portal.

Step 2: Initial Investigation

Investigation teams shall determine whether the incident amounts to a cybercrime.

Step 3: Warrant Application

Authorities shall seek warrants from appropriate courts before searching any private property or seizing any communications.

Step 4: Digital Evidence Gathering

Digital forensics specialists will gather, secure, analyze, and document evidence in a process that ensures proper chain of custody of evidence.

Step 5: Case Referral

Sufficient cases shall then be referred to dedicated cyber courts for prosecution. All investigations shall be undertaken within the rule of law.

International Coordination

Cyber threats tend to emanate from beyond a country's borders. Consequently, the government of VIT's will need to collaborate with international cybersecurity organizations, other governments, and police services in addressing cybercrimes.

VIT's will need to:

- * Exchange information on cyber threats with their partners.
- * Join international cybersecurity drills.
- * Conduct investigations into cybercrime cases across borders.
- * Swap digital evidence with other countries legally.
- * Facilitate extradition processes for serious cybercriminals.
- * Respond to cyber emergencies worldwide.

The NCSA will need to communicate directly with international organizations like INTERPOL, Regional CERTs, and International Cybersecurity Alliances.

International collaboration programs will be conducted in line with national privacy legislation and international law.

Awareness Programs

The Government of VIT's shall conduct nationwide cybersecurity awareness programs to educate citizens, businesses, and public institutions about cyber threats and safe online practices.

Public awareness initiatives shall include:

- **School and University Programs**
Cybersecurity education shall be integrated into school and university curricula to teach students about online safety, digital privacy, password security, and cyber ethics.
- **Public Awareness Campaigns**
The government shall conduct regular awareness campaigns through television, radio, newspapers, websites, and social media platforms.
- **National Cybersecurity Awareness Month**
A dedicated annual campaign shall be organized to promote cybersecurity best practices among citizens and organizations.
- **Business Training Programs**
Workshops and training sessions shall be provided to businesses, especially small and medium enterprises (SMEs), to strengthen cybersecurity preparedness.
- **Cyber Emergency Alerts**
Citizens shall receive official alerts regarding significant cyber threats, scams, phishing campaigns, and emerging vulnerabilities through government communication channels.

The effectiveness of awareness programs shall be reviewed annually, and improvements shall be implemented based on emerging cyber threats and public feedback.

Enforcement and Accountability

All Government Agencies must adhere to the cybersecurity regulations established by the National Cyber Security Authority (NCSA). In the event that any agency does not apply the necessary cybersecurity measures, internal audits and penalties can be enforced against them.

The NCSA will issue the Annual National Cybersecurity Report to Parliament highlighting:

- * The major cyber threats that affect VITs.
- * Statistics for cybercrimes.
- * Security status of critical infrastructure.
- * Incident response effectiveness.
- * Awareness campaigns.
- * Recommendations for future cybersecurity.

PART 8: Cyber Incident Reporting Framework

The Cyber Incident Reporting Framework establishes a standardized process for detecting, reporting, investigating, responding to, and recovering from cybersecurity incidents within VIT's. All government agencies, businesses, educational institutions, healthcare organizations, and critical infrastructure operators shall comply with this framework.

Reporting Process Steps

Step	Stage	Description (Student to Complete)
1	Incident Detected	A cybersecurity event is detected via system monitoring, employee notification, client feedback, security warnings, or through detection tools. The victimized company has to form an Incident Response Team (IRT) and collect all available information. The critical incident, involving sensitive data or critical infrastructure, should be escalated in *1 hour*.
2	Internal Investigation	An initial investigation to determine the nature and extent of the incident must be performed by the organization's Incident Response Team. Evidence, which includes system logs, network data, user activity, and impacted files, must be secured. The evaluation process is to be completed within *24 hours* after discovery of the incident.
3	Report to Authority	In cases where the breach relates to any of the following: unauthorised access, data breach, ransomware attacks, malware infections, financial fraud or infrastructure disruptions, then it is mandatory for the organisation to report the matter to the *National Cyber Security Authority (NCSA)* within *72 hours*.
4	Risk Assessment	The National Cyber Security Agency will work in conjunction with the victimized institution to assess the level of damage caused. The factors to consider will include the number of affected citizens, the kind of information compromised, financial loss incurred, impact on the provision of public services, and any threats posed to national security.
5	Mitigation	Proper containment efforts should be carried out to prevent further damage. This may entail such actions as system isolation, disabling of the compromised accounts, blocking of the malicious network traffic, application of security patches, etc. NCSA may give out mandates for certain types of situations.
6	Recovery	Restoration of systems, services, and information shall be done using backups that have been tested and recovery procedures that are authorized. Security testing will be performed before restoration to normal operations of systems. Any citizen who has been affected will be informed in case of any danger to their personal identification, finances, and privacy.
7	Final Report	The organization is required to present the Final Incident Report to the NCSA within *30 days* of resolving the incident. This report should contain the findings from the root cause analysis and measures adopted in response to the incident and the lessons learned.



Roles & Responsibilities

Organizations

- * Detect & report cyber incidents.
- * Preserve digital evidence.
- * Assist during investigations.
- * Mitigate and recover from the situation.
- * Inform affected parties as appropriate.

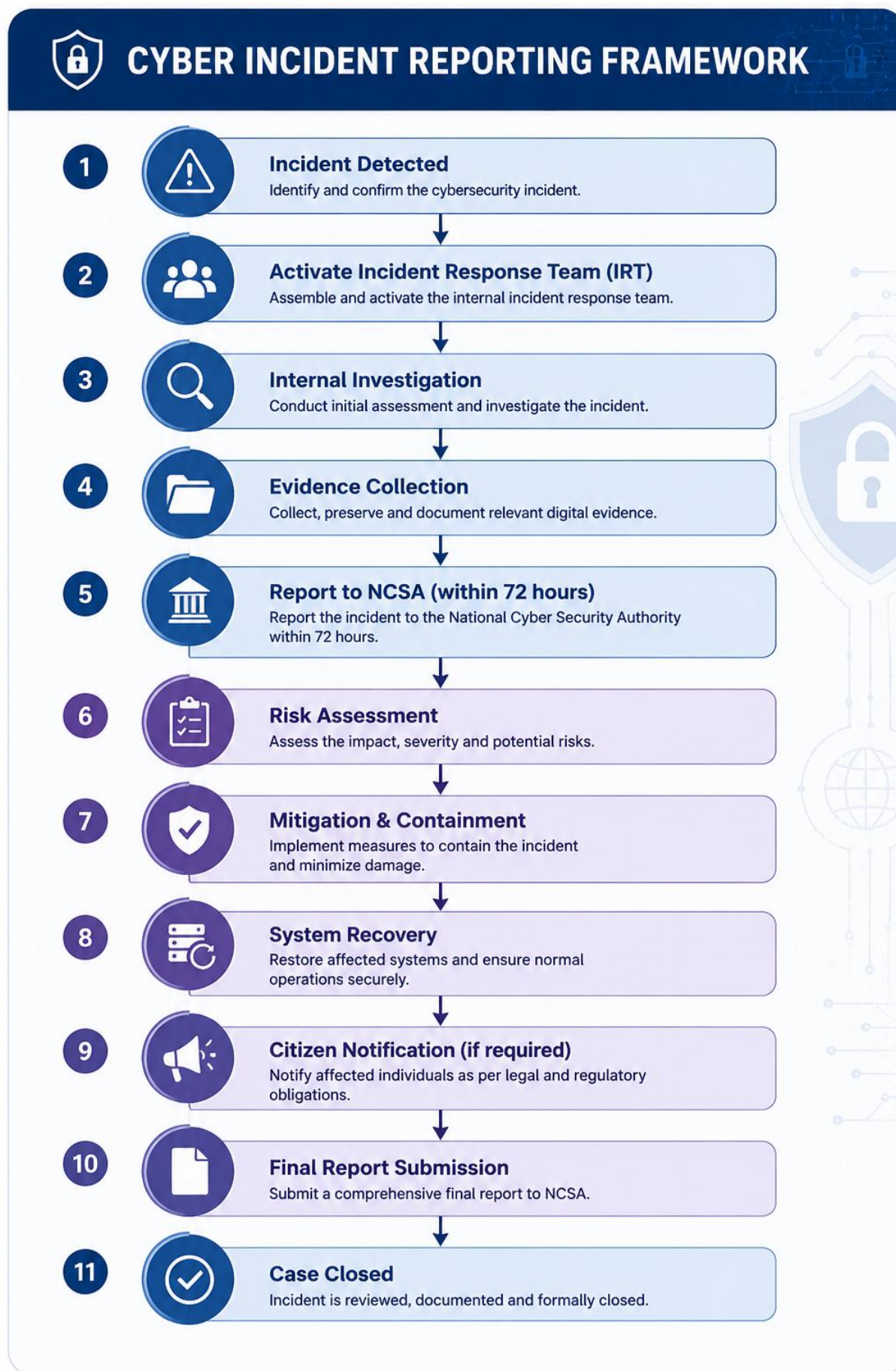
National Cyber Security Authority (NCSA)

- * Receive incident reports.
- * Facilitate the national response.
- * Carry out risk assessment.
- * Offer technical advice.
- * Investigate large-scale incidents.
- * Ensure reporting is in compliance.

Law Enforcement Agencies

- * Investigate cyber crimes.
- * Gather and preserve forensic evidence.
- * Identify suspects for prosecution.
- * Coordinate with international organizations as required.

Flowchart: Cybercrime Reporting



PART 9: National Cyber Security Authority (NCSA)

The National Cyber Security Authority (NCSA) shall be the central government agency responsible for protecting the digital infrastructure, citizens, businesses, and national interests of VIT's from cyber threats. The NCSA shall operate under the Ministry of Digital Security and shall possess regulatory, investigative, operational, and advisory powers.

Organizational Structure

The NCSA will have at its helm a *Director General (DG)* who is appointed by the Government of VIT's for a term of five years.

NCSA Leadership Hierarchy:

1. Director General (DG)
 - Head of the NCSA.
 - In charge of national cyber strategy.
 - Reports to the Minister of Digital Security.
2. Deputy Director General - Operations
 - In charge of incident response and cyber defense operations.
 - Coordinating national emergency responses.
3. Deputy Director General – Investigations
 - In charge of cybercrime investigations and digital forensics.
4. Deputy Director General – Compliance & Regulation
 - In charge of inspections and other enforcement actions.
5. Deputy Director General – Public Awareness & Education
 - In charge of public awareness initiatives and cybersecurity training programs.

NCSA Reporting Structure:

Director General → Deputy Directors General → Department Directors → Regional Cybersecurity Offices → Cybersecurity Officers and Analysts

NCSA offices will be set up in different regions of VIT's to ensure quick response to any cyber threats within the country.

Core Responsibilities

The NCSA will have the following main duties and responsibilities:

National Cybersecurity Policy Development, revision, and implementation of national cybersecurity policies, regulations, and standards.

1. **Cyber Threat Monitoring:** Detection of cyber threats affecting citizens, private businesses, public institutions, and critical infrastructure.
2. **Cyber Incident Response Coordination:** Coordinating the national response for cyber attacks and cyber emergencies.
3. **Cybercrime Investigation Assistance:** Provide technical assistance to law enforcement agencies in investigating cybercrimes.
4. **Compliance with Regulations:** Inspections and auditing to ensure compliance with cyber-related laws.
5. **Critical Infrastructure Protection:** Ensure the security of banking systems, hospitals, power generation systems, transportation systems, and government networks.
6. **Cybersecurity Awareness Programs:** Organize awareness programs about cybersecurity and training for citizens on cybersecurity issues.
7. **International Cooperation:** Work with cybersecurity agencies, CERTs, and other organizations from other countries to tackle cybercrime.
8. **Cybersecurity Research:** Encourage the research of new cyber-related technology and artificial intelligence security, quantum encryption technology.

Legal Powers

For effective discharge of its functions, the NCSA should have the following legal powers:

1. Regulatory Powers
 - Develop cybersecurity rules and regulations.
 - Make requirements for implementation of minimum security measures.
 - Conduct compliance checks and audits.
2. Information Gathering Powers
 - Require organizations to provide cybersecurity information.
 - Require submission of incident reports, security audit reports, and technical documents.
 - Compel organizations to submit information in support of investigations.
3. Emergency Powers
 - Issue emergency security directions in case of cyber emergencies within the nation.
 - Direct isolation of infected systems temporarily.
 - Mobilize and coordinate emergency incident response.
4. Website and Service Blocking Powers
 - Block harmful websites.
 - Terminate fraudulent online services.
 - Prevent access to cyber criminal infrastructure.

5. Administrative Fines

- The NCSA has the authority to impose administrative penalties for violation of cybersecurity regulations.

6. Referral Powers

- Cases involving serious offenses are referred to appropriate agencies and courts of cybercrime for criminal proceedings.

Investigation Process

The NCSA will adhere to a standard process for conducting cybercrime investigations:

Step 1: Complaint Registration

The complaints will be received from citizens, businesses, government agencies, and foreign partners using the National Cyber Crime Portal, Emergency Hotline, and regional NCSA offices.

Step 2: Preliminary Evaluation

The nature, degree, impact and jurisdiction of the incident are evaluated. Cases are ranked as either Low, Medium, High, or Critical risk.

Step 3: Evidence Collection and Analysis

Investigators collect and secure the relevant evidence. This includes log files, network activities, communications, malware samples and hard drive images in accordance with digital forensics procedures. The Digital Forensics Department analyses the evidence, reconstructs lost data, determines the source of attacks, and finds the indicators of compromise.

Step 4: Identification of Suspects and Enforcement Actions

Individual perpetrators, organized crime, insiders or foreign agents behind the attack are identified. International coordination can be undertaken where necessary. Depending on the findings, the NCSA may give out warnings, impose penalties, make referrals, obtain legal warrants and emergency action.

Step 5: Conclusion and Reporting of the Case

The Final Investigation Report documents the case, findings and actions taken. All major cases should be documented in the National Cyber Incident Database.

Departments

The following specialized departments should be established within the NCSA:

1. Threat Intelligence Unit (TIU)

Functions:

- * Watch global cyber threats.
- * Gather threat intelligence.
- * Investigate attacks.
- * Issue threat advisories.

2. National Incident Response Team (NIRT)



Functions:

- * React to cyber crises.
- * Facilitate national incident response.
- * Assist affected organizations.
- * Deal with cyber crises situations.

3. Digital Forensic Laboratory (DFL)

Functions:

- * Recover digital evidence.
- * Analyze cybercrime artifacts.
- * Perform forensic investigation.
- * Help prosecutions.

4. Compliance and Audit Department (CAD)

Functions:

- * Carry out security audits.
- * Monitor regulation compliance.
- * Inspect critical infrastructure operators.
- * Suggest enforcement actions.

5. Public Awareness and Education Division (PAED)

Functions:

- * Run cybersecurity awareness campaigns.
- * Create training programs.
- * Publish educational material.
- * Encourage cyber hygiene practices.

6. Critical Infrastructure Protection Division (CIPD)

Functions:

- * Protect vital national services.
- * Evaluate infrastructure risks.
- * Test for resilience.
- * Facilitate infrastructure protection programs.

7. International Cooperation Office (ICO)

Functions:

- * Coordinate with international bodies.
- * Exchange cyber threat intelligence.
- * Assist cross-border investigations.
- * Contribute to international cybersecurity programs.

PART 10: AI and Emerging Technology Regulations

With the increase in the incorporation of emerging technologies into our society, the Government of VIT's shall put in place cybersecurity laws to make sure that they are safely, securely, and responsibly used. The following are the regulations concerning emerging technologies in VIT's.

Artificial Intelligence

Key Risks

Risks that Artificial Intelligence can bring from a cybersecurity perspective include:

- Attacks using phishing and social engineering methods through AI.
- Breach of AI training data sets.
- Algorithm tampering and data poisoning attacks.
- Bias and wrong decision-making in AI systems.
- Theft of AI algorithms and intellectual property.
- Cyberattacks automated by AI.
- Violations of privacy due to over-collection of data.

Regulations

1. Cybersecurity and risk assessment must be conducted for all AI systems which present a risk, whether in the fields of healthcare, banking, law enforcement, education, and government services, etc.
2. Training data, model updates, and security testing details should be documented by the developers of AI systems.
3. Data poisoning, model theft, and unauthorized access to AI systems should be prevented.
4. Citizens will have the right to ask for explanation of the important decisions made by AI.
5. AI cannot be used for illegal monitoring of citizens.
6. Fines up to ₹5,000,000 can be imposed on organizations that fail to meet AI security guidelines.

Deepfakes

Key Risks

Some of the issues that can be experienced by Deepfakes include:

- Identity theft and fraud.
- Interference with election campaigns.
- Cyberfraud and social engineering.
- Destruction of one's reputation through fake media.
- Dissemination of misleading information.
- Harassment.

Regulations

1. All the audio, visual and imagery generated by AI should have a watermark that signifies that it is AI-generated.
2. Malicious deepfakes made for the purpose of fraud, harassment, rigging of elections and any form of crime will be prohibited.
3. Any person whose images or voices appear in AI-generated imagery should give permission to do so unless the content is satirical, educational or news-based.
4. The malicious deepfakes on social media sites need to be removed within 24 hours after receiving an official complaint.
5. Organizations using AI-generated imagery should keep track of who creates such content.
6. People violating this law by making malicious deepfakes will face imprisonment and monetary penalties.

IoT Devices

Key Risks

IoT technology may put citizens and organizations at risk of:

- Weak default passwords.
- Unauthorized remote access.
- Data leakage.
- Hijacking of devices and botnets.
- Surveillance via infected devices.
- Insecure software updates.

Regulations

1. All IoT devices to be sold inside VIT's jurisdiction must have individual default credentials and secure authentication.
2. There must be security updates for at least five years from the time the products are made available.
3. The IoT devices must have the capacity to communicate securely and install security updates.
4. Critical IoT devices that will be deployed in healthcare, transportation, and energy must pass through security certification.
5. Disclosure of the vulnerabilities and provision of security patches must be done by manufacturers.
6. Selling IoT devices with critical security issues is prohibited.

Cloud Computing

Key Risks

Cloud environments could encounter:

- Data breaches.
- Inadequately configured cloud storage solutions.
- Internal threats.
- Unauthorized access.
- Service disruption.
- International transfer of data risks.

Regulations

1. The cloud services should provide encryption in transit and at rest.
2. Disaster Recovery & Business Continuity should be provided by the service providers.
3. Cloud service users should perform annual audits and risk assessment.
4. The transfer of sensitive citizen information outside VITs needs clearance from NCSA.
5. The cloud services should report any security incidents to the NCSA within 72 hours.
6. Multi-Factor Authentication should be required for all administrative accounts.
7. Failing to secure user data might lead to huge fines and license cancellation.

Digital Identity Systems

Key Risks

Digital ID mechanisms are susceptible to:

- Identity fraud.
- Credential theft.
- Breach of biometric data.
- Unauthorised access to government services.
- Massive privacy breaches.
- Account takeover attacks.

Regulations

1. Multi-factor authentication will be required for all digital identity schemes.
2. Biometrics will be encrypted and kept apart from personal identity information.
3. Citizens will have the right to review and correct any incorrect identity data.
4. Annual security reviews of digital identity systems will be conducted by government organizations.
5. Disclosure of any digital identity information will be reported to citizens and NCSA within an immediate timeframe.
6. Misuse or sale of digital identity information is considered a serious cybercrime.

Autonomous Systems

Key Risks

In the case of autonomous systems including self-driving cars, drones, robotics, and automation of industrial machinery, possible threats include:

- Remote hijacking.
- Manipulation of software.
- GPS Spoofing attacks.
- Safety hazard due to cyberattacks.

Regulations

1. All autonomous systems should be certified through cybersecurity before their deployment.
2. The manufacturers should ensure that there are secure update software processes and anti-tampering capabilities.
3. Autonomous systems should keep logs of their activities for analysis purposes.
4. Critical autonomous systems should have an emergency manual override capability.
5. Organizations using autonomous systems should conduct penetration tests annually.
6. Cyber incidents involving autonomous systems should be reported to the NCSA within 24 hours.
7. Manufacturers could face liability for any harm caused by negligent cybersecurity practices.

PART 11: Critical Infrastructure Protection

Critical infrastructure sectors are essential for national security, public safety, economic stability, and the well-being of citizens. The following sector-specific cybersecurity requirements shall be mandatory for all critical infrastructure operators within VIT's.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	1. Multi-Factor Authentication (MFA) shall be enabled for both customers and employees' accounts. 2. All financial transactions and customer data shall be encrypted using end-to-end encryption. 3. Fraud detection system shall be used for monitoring any suspicious transactions and activities in customer accounts. 4. An annual penetration test and security audit shall be carried out and the report sent to the NCSA. 5. Disaster recovery and back-up systems shall restore all services within 24 hours of a cyber attack.
Hospitals & Healthcare	1. EHRs shall be encrypted at rest and in transit. 2. Networked medical devices shall be subject to cybersecurity certification prior to deployment. 3. Access to patient information should be restricted through role-based controls in relation to the tasks of the job. 4. Hospitals should have secure offline backups for patients' records to defend against ransomware attacks. 5. Cybersecurity breaches that impact patients' information should be reported within 24 hours to the NCSA by healthcare organizations.
Airports & Aviation	1. Network segregation will ensure that air traffic control systems are separated from any public or administrative networks. 2. Monitoring systems should constantly monitor any unauthorized access and abnormal network activity. 3. Multi-level authentication process must be applied to critical aviation systems. 4. Cybersecurity and incident response exercises must be performed on an annual basis. 5. Redundant communications system should be maintained by aviation operators.
Railways & Transport	1. Signaling, scheduling, and traffic control systems need to be secured using network isolation and access control mechanisms. 2. Real-time monitoring systems should always be able to detect any cyber attacks within the transportation networks. 3. All the transport management software needs to be updated regularly. 4. The emergency response plans should be tested once per year through cyber attack simulations.



	5. All the critical transportation systems should have alternate methods for operation.
Power Plants & Energy	1. The ICS and SCADA networks should be isolated from the company's network. 2. Security monitoring should be performed continuously in all OT environments. 3. Multi-factor authentication should be used when accessing the electricity production and distribution systems. 4. Quarterly vulnerability assessments and annual penetration testing should be conducted in critical energy infrastructure. 5. Backup control centers and emergency systems should be kept to provide uninterrupted power during a cyberattack.
Government Data Centers	1. Government information must be encrypted during storage and transmission according to recognized encryption standards. 2. Access to government databases will be limited via role-based access control and multi-factor authentication. 3. SIEM tools must be used to monitor network traffic and security log files continuously. 4. Government data centers will have redundant sites located in distant geographical locations for disaster recovery purposes. 5. Yearly security audits, penetration tests, and compliance checks will be compulsory under NCSA oversight.

PART 12: Awareness and Education

National Cybersecurity Awareness Campaign: "Secure VIT's 2035"

The Government of VIT's, through the National Cyber Security Authority (NCSA), shall implement a nationwide cybersecurity awareness campaign called **"Secure VIT's 2035."** The campaign aims to educate citizens of all age groups about cyber threats, safe online behavior, privacy protection, and responsible use of technology.

The campaign shall focus on five major target groups: school students, university students, employees, social media users, and the general public.

School Programs (Age 6-18)

The Ministry of Education shall integrate cybersecurity education into the national school curriculum.

Age Group 6–10 (Primary School)

Topics: Safe internet use, personal information protection, strong passwords, and cyberbullying awareness.

Activities: Cyber safety games, educational videos, quizzes, and storytelling sessions.

Outcomes: Students will learn safe internet practices, basic privacy concepts, and how to report suspicious online activities.

Age Group 11–14 (Middle School)

Topics: Social media safety, phishing awareness, password management, digital footprints, and online scams.

Activities: Cybersecurity competitions, poster creation, role-playing exercises, and interactive online learning.

Outcomes: Students will identify phishing attempts, secure personal accounts, and practice responsible digital citizenship.

Age Group 15–18 (High School)

Topics: Cyber ethics, privacy protection, cybersecurity careers, ethical hacking concepts, and cybercrime laws.

Activities: Cybersecurity clubs, CTF competitions, Cyber Safety Olympiads, and expert guest lectures.

Outcomes: Students will understand cyber threats, apply cybersecurity best practices, explore career opportunities, and demonstrate responsible online behavior.

College & University Programs

Higher education institutions shall implement formal cybersecurity education programs.

Cybersecurity Educational Courses

All undergraduate students shall be required to undertake cybersecurity courses including:

- Basics of network security
- Cyber laws and ethics
- Privacy
- Incident handling
- Artificial Intelligence Security
- Cloud Security
- Digital Forensics

Certification Tracks

Higher education institutions shall offer the option of pursuing the following:

- Cybersecurity Foundation Certification
- Ethical Hacking Certification
- Network Security Certification
- Cloud Security Certification
- Digital Forensics Certification

Lab Facilities

Higher education institutions shall ensure that there are:

- Cybersecurity laboratories
- Virtual cyberspace ranges
- Ethical hacking lab facilities
- Malware laboratory facilities
- Digital forensics laboratory facilities

Student Learning Outcomes

Students will:

- Understand the basics of cybersecurity.
- Acquire cybersecurity skills.
- Engage in research work.
- Prepare for industry certifications and jobs.

Corporate Training Programs

All organizations in the VIT's jurisdiction must undergo mandatory cybersecurity training for employees.

Training Requirements

1. Frequency
 - Employee New Hire Training: Within 30 days of employment.
 - Refreshing Training: Twice annually.
 - Specialist Training: Once annually for IT and security staff.
2. Training Subjects
 - Password security
 - MFA
 - Phishing protection
 - Social engineering detection
 - Data protection policies
 - Incident reporting procedures
 - Security while telecommuting

3. Evaluation Techniques

Organizations must implement:

- Online tests
- Simulated phishing tests
- Cybersecurity simulations
- Security practices

Employees will have to pass at least 80% on the test.

4. Consequences of Failure to Provide Mandatory Training
 - The NCSA can impose sanctions on organizations that fail to provide mandatory training.

Social Media Campaigns

The NCSA will run campaigns on social media to make people aware of various aspects of cybersecurity.

Platforms:

The campaign will run on:

- Instagram
- Facebook
- YouTube
- LinkedIn
- X (formerly Twitter)
- TikTok

Themes of the Campaign:

- Think Before You Click
- Protect Your Password
- Stop Phishing Attacks
- Protect Your Digital Identity

- Spot Deepfakes
- Protect Your Children Online
- Report Cybercrime

Types of Content:

- Educational videos
- Explain videos
- Infographics
- Interactive quiz
- Cybersecurity tips
- Live question-and-answer sessions
- Cases of cybercrimes

Public Awareness Workshops

The government will conduct free community cybersecurity workshops in all VITs.

1. Venue

The workshops will be held in:

- Schools
- Colleges
- Community centers
- Libraries
- Government offices
- Rural digital service centers

2. Frequency

- Urban Areas: Monthly workshops
- Rural Areas: Quarterly workshops
- National Cybersecurity Week: Annual intensive awareness drive

3. Beneficiaries

- Senior citizens
- Parents
- Small businessmen
- Teachers
- Government officials
- Rural residents

4. Workshop Contents

- Online banking safety measures
- Security of mobile phones

- Protection against cyber frauds
- Privacy in cyberspace
- Proper use of social media
- Cybercrime reporting methods

5. Modes of Conducting Workshops

- Face-to-face seminars
- Demonstrations
- Practical learning
- Distribution of education leaflets
- Online webinars

PART 13: International Cooperation

Cybercrime often involves crossing national boundaries, thus requiring international cooperation for cybersecurity measures. The Government of VIT's must work together with foreign governments, international bodies, law enforcement, and cybersecurity entities in order to combat the problems associated with cybercrimes. NCSA is the main government organization responsible for conducting international cybersecurity cooperation efforts.

The Government of VIT's will embrace similar principles as other internationally recognized principles of cybercrime like the Budapest Convention on Cybercrime without violating national, privacy, and human rights requirements.

Cybercrime Investigations

Cyber criminals may work in more than one country, hence the need for international cooperation when conducting such investigations.

1. Policy

The government of VIT's will cooperate with foreign law enforcement agencies in investigating cybercrime that includes the following:

- Cross-border cyberattacks
- International ransomware
- Financial crimes on the Internet
- Identity fraud
- Crimes involving child abuse
- Cyber terrorism
- Infrastructure attacks

2. Procedure

- The NCSA will create specialized International Cybercrime Investigation Units.
- The request for assistance will be made through diplomatic routes, Mutual Legal Assistance Treaties (MLAT), or cybersecurity cooperation agreements.
- Digital evidence to be acquired from foreign countries will be gathered in accordance with internationally recognized forensics standards.
- Joint investigation teams may be set up in cases where there is international cybercrime.
- The information received through international investigations will be used only for law enforcement purposes.

3. Objective

In order to prevent cyber criminals from evading justice using the fact that they are operating in more than one country.

Information Sharing Agreements

Timely information sharing is essential to detect and prevent cyber threats before significant damage occurs.

1. Policy

VIT's shall enter into bilateral and multilateral cybersecurity information-sharing agreements with trusted partner nations and international organizations.

2. Procedures

- The NCSA shall establish secure communication channels with foreign cybersecurity agencies.
- Information sharing may include:
 - Malware indicators
 - Threat intelligence reports
 - Vulnerability information
 - Attack methodologies
 - Cyber incident reports
 - Risk assessments
- Shared information shall be classified according to sensitivity levels and protected using appropriate security controls.
- Personal information shall only be shared when legally authorized and necessary for cybersecurity purposes.
- Participating organizations must comply with data protection and privacy regulations.

3. Objective

To improve early warning capabilities and strengthen collective cyber defense against emerging threats.

Joint Operations

Many instances of massive cyber threats need an international response strategy.

1. Policy

The Government of VIT's shall engage in joint cybersecurity actions alongside other international parties for disrupting the cyber criminals' activities and ensuring global security in cyberspace.

2. Procedures

- Joint actions can include:
 - Cyber crime investigation

- Disruption of ransomware attacks
 - Botnets takedown actions
 - Analysis of malware
 - Critical infrastructure security
- The NCSA shall work with:
 - Foreign CERTs
 - International law enforcement agencies
 - Cybersecurity researchers
 - International cybersecurity coalitions
- Agencies participating in joint actions shall establish:
 - Scope of actions and responsibilities of agencies
 - Information exchange policy
 - Policies on handling of evidence
 - Objective of actions
- Joint actions shall be carried out within the scope of national and international agreements.

3. Objective

To enhance the global cybersecurity by coordinating actions against cyber criminal networks.

Extradition Requests

Cybercriminals frequently operate from foreign jurisdictions to avoid prosecution.

1. Policy

The Government of VIT's shall seek the extradition of individuals accused of serious cybercrimes when sufficient evidence exists and legal requirements are satisfied.

2. Procedures

- Extradition requests shall be submitted through established diplomatic and legal channels.
- Extradition may be requested for offenses including:
 - Cyber terrorism
 - Large-scale ransomware attacks
 - Financial cyber fraud
 - Critical infrastructure attacks
 - Organized cybercrime
- The requesting authority must provide:

- Evidence supporting the charges
- Details of the alleged offense
- Applicable legal provisions
- Information regarding penalties
- Extradition requests shall respect:
 - Human rights protections
 - Due process requirements
 - International legal obligations

3. Objective

To prevent cybercriminals from escaping justice by relocating to foreign countries.

Threat Intelligence Exchange

Since cyber threats change quickly, there is a need for continual intelligence sharing between countries.

1. Policy

There should be an International Threat Intelligence Exchange Program set up by the NCSA.

2. Procedures

- The NCSA shall share intelligence on threats with:
 - Foreign cybersecurity agencies
 - International CERTs
 - Regional security alliances
 - Cybersecurity organizations globally
- The intelligence shared can be:
 - Indicators of Compromise (IOCs)
 - Malware signatures
 - Vulnerabilities that have just been discovered
 - Threat actors' profiles
 - Attack trends
 - Security advisories
- Threat intelligence platforms will help increase the speed of intelligence sharing.



- Information shared must be verified prior to its distribution to the operators of critical infrastructures.
- Sensitive intelligence information will be safeguarded via secure communication channels.

3. Objective

To improve national and international preparation against changing cyber threats.

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: The National University Data Breach

Crime Category	Unauthorized Access / Data Breach
What Happened	March 2034 saw an ex-IT contractor for the National University of VIT take advantage of the university's administrative account which had not yet been terminated following expiration of his employment period. Utilizing the access credentials to this account, the individual proceeded to illegally gain access to the institution's students' information system. The individual then proceeded to steal the personal information of about 50,000 students of the university. The information included such things as their names, addresses, academic records, identification numbers, among others. The data was eventually sold in the underground marketplaces for cybercrimes. This breach was discovered following suspicious activities observed within the database.
Law Violated	The violation committed by the perpetrator includes the following: Section 4: Unauthorized Access Section 4: Data Theft Part 5: Data Protection Policy Part 6: Corporate Security Compliance Requirement Part 8: Cybersecurity Incident Reporting Requirement
Investigation Process	The university promptly informed the NCSA about the incident. Activities performed by the NCSA: 1. Collection of server logs and access logs. 2. Digital forensics investigation. 3. Identification of login attempts made from the address of the ex-contractor. 4. Acquisition of a court warrant for confiscation of digital devices. 5. Recovery of the stolen data from the suspect's computer. 6. Finding evidence that shows an attempt to sell the stolen data. The suspect was apprehended and charged according to the National Cybersecurity Law.
Punishment Given	The offender was sentenced to: * 6 years of imprisonment. * Fine of ₹1,500,000.



	* Permanently barred from serving in sensitive IT jobs. * Confiscation of any proceeds from the crime committed.
Lessons Learned	<i>Post-incident Actions:</i> * Implementation of mandatory deactivation of user account protocols. * Quaterly access checks were mandated by universities. * Multi-Factor Authentication (MFA) was made mandatory for administrator accounts. * Offboarding process for employees was improved.

Case Study 2: Operation FakeBank

Crime Category	Financial Fraud / Phishing
What Happened	<i>[In August 2034, hackers initiated a phishing scam against clients of the National Bank of VIT's.</i> <i>These customers were sent emails that informed them about the suspension of their bank accounts unless they updated their details. This email was linked to a fake website that ooked exactly like the real website.</i> <i>Over 12,000 individuals provided their usernames, passwords, and one-time passwords (OTPs). Hackers made use of this data to move money out of the accounts and into cryptocurrency wallets.</i> <i>This hack resulted in the loss of more than ₹80,000,000.</i>
Law Violated	<i>Violations made by the offenders include the following:</i> * Section 4: Phishing * Section 4: Financial Fraud * Section 4: Identity Theft * Section 10: AI-Fraud Prevention Regulations * Part 8: Incident Reporting Procedures
Investigation Process	<i>This was notified to the NCSA by the National Bank within 24 hours.</i> <i>Actions taken by NCSA:</i> <i>1. Collection of phishing emails and fake website information.</i> <i>2. Trace of domain registration information.</i> <i>3. Analysis of cryptocurrency transactions.</i> <i>4. Collaboration with cybersecurity agencies in other countries.</i> <i>5. Identifying the criminals from the criminal groups in various countries.</i> <i>6. Joint international operations that led to several arrests.</i> <i>Digital evidence collected includes email servers, payments,</i>



	cryptocurrency wallets, and other seized machines.
Punishment Given	The court punished the principal culprits with: * Imprisonment for ten years. * Fine of ₹3,000,000 each. * Restitution of all stolen money. * Ban on conducting any financial activity forever.
Lessons Learned	Post the occurrence, * The banks used AI-based phishing detection software. * Phishing prevention awareness programs at a national level were enhanced. * Monthly phishing drills became mandatory for financial institutions. * Transaction monitoring software was updated.

Case Study 3: Blackout Storm

Crime Category	Ransomware / Cyber Terrorism
What Happened	The first cyberattack took place on the National Power Grid of VIT's in January 2035 by an experienced gang of cybercriminals. Attackers managed to find vulnerabilities in industrial control systems (ICS), which are linked to regional power plants. Hackers planted malware in the operational technology network, leading to disruption of power supplies in multiple large cities. This attack affected approximately 2 million citizens and caused a total loss of ₹500,000,000 due to the shutdown of transportation, hospitals, governmental departments, and economic processes. The attack is considered to be a national cyberemergency.
Law Violated	The offenders violated: * Section 4: Cyber Terrorism * Section 4: Unauthorized Access * Section 4: Malware Distribution * Part 11: Critical Infrastructure Protection Requirements * Part 10: Autonomous and Industrial Systems Regulations
Investigation Process	The NCSA implemented the National Cyber Emergency Response Framework. These included: 1. Isolation of systems under attack immediately. 2. Activation of the National Incident Response Team. 3. Analysis of malware by the Digital Forensics Laboratory.



	4. Threat intelligence sharing across the globe. 5. Investigation of the Command-and-Control (C&C) infrastructure. 6. Cooperation with international law enforcement authorities. 7. Capture of major criminals by multinational coordinated operations. Investigation found that the malware was created by a cybercrime organization attacking energy infrastructures.
Punishment Given	Judgment of the Cyber Crime Court was that the act was cyber terrorism. The ringleaders were punished with the following: - 25 years of jail term. - Fined ₹10,000,000 each. - Confiscation of assets. - Lifelong ban on handling any information system.
Lessons Learned	Post-attack measures included: * Network segmentation was mandated for all power plants. * Security certification was made mandatory for industrial control systems. * Vulnerability assessments had to be done quarterly. * Annual national cyber emergency exercises were initiated. * Backup control centers were set up throughout the country.

PART 15: Future Recommendations

With the rapid development of technology, the Cybersecurity Law of VIT's needs to change in order to combat any new challenges and opportunities. The list below gives suggestions about future changes that should be implemented into the national framework within the next 10 years.

AI Security (2025-2035)

Artificial Intelligence is going to become even more advanced and able to make decisions on its own. Although there are some positive aspects of using Artificial Intelligence, new cybersecurity threats will appear.

1. Future Challenges

- AI-driven cyberattacks.
- Automated phishing and social engineering attacks.
- Deepfakes creation.
- Bias in AI decision-making.
- Autonomous AI operation without human supervision.
- Manipulation of people's opinions and votes by AI systems.

2. Recommendations

- Annual Security Audits of AI Systems
All critical AI systems should have annual cybersecurity and ethical audits prior to their deployment.
- Human Supervision of Critical AI Decisions
Important decisions made by AI systems in healthcare, law enforcement, finance, and governmental services shall be reviewed by humans.
- AI System Documentation
Companies shall provide documentation about:
 - Data sources for model training.
 - Behavior of the model.
 - Security audit findings.
 - Decision-making process.
- AI Threat Intelligence Programs
The NCSA shall develop dedicated teams for finding and defending against AI-driven cyberattacks.
- National AI Safety Certification
Government shall introduce certification process for critical AI systems before public deployment.

Quantum Computing

Many current encryption schemes may be rendered useless if quantum computers are developed in the future.

1. Future Threats

- The decryption of currently secure communications.
- The exposure of government and military data.
- Dangers to the banking system and other financial institutions.
- Persistent threat to data stored in an encrypted form.

2. Recommendations

- Quantum Readiness Initiative
The Government of VIT's should develop a National Quantum Security Program.
- Post-Quantum Cryptography
Operators of critical infrastructure will start migrating towards post-quantum cryptography by 2030.
- Periodic Security Audits
Firms and organizations working with sensitive data should perform quantum readiness audits once every two years.
- Investments in Research
The government will provide funding for the development of:
 - Quantum-safe encryption.
 - Quantum key distribution.
 - Advanced cryptographic techniques.
- National Quantum Testing Center
The NCSA will set up a facility for testing quantum security products and solutions.

Privacy Protection

As more personal data is being collected by means of digital technologies, the need for better privacy regulations will arise.

1. Future Issues

- Biometric surveillance.
- Behavioral profiling.
- Smart devices monitoring.
- Tracking of locations.
- Personal data analysis through AI.
- Data aggregation.

2. Recommendations

- Improved Biometric Data Protection
Biometric data shall enjoy the maximum level of legal and technical protection and encryption.



- **Right to Remain Anonymous Online**
Citizens shall have the right to use some online services anonymously without personal information being collected.
- **Limitations on Behavioral Profiling**
It shall be forbidden to create behavioral profiles without consent.
- **Privacy Features for Smart Devices**
Manufacturers shall enable users to control data collection and sharing on their smart devices.
- **Extended Right to be Forgotten**
Citizens shall have the right to delete unnecessary or outdated personal information

Digital Identity

Systems for secure digital identity will be necessary for the delivery of future government and business services.

1. Future Challenges

- Identity Theft.
- Breach of the centralized database.
- Compromise of biometric information.
- Unsanctioned surveillance.
- Exclusion from digital platforms by vulnerable groups.

2. Recommendations

- **Decentralized Identity System**
VIT's will implement a decentralized digital identity system to minimize the use of centralized databases.
- **Multi-factor Authentication**
All digital identity systems must incorporate multi-factor authentication systems.
- **Control by Citizens over their Information**
Citizens will know:
 - Who has access to their information.
 - What purpose is served by accessing their information.
 - The time of such access.
- **Privacy Technology Solutions**
Digital identities will adopt:
 - Encryption technology.
 - Zero-Knowledge proof technology.
 - Secure authentication processes.
- **Alternative Identification Services**
Identification methods shall be available for those citizens who lack access to advanced technology systems.



Cyber warfare is expected to become a major national security concern over the next decade.

1. Future Challenges

- State-sponsored cyber attacks.
- Attacks on critical infrastructure.
- Cyber espionage.
- Information warfare and disinformation operations.
- Essential services disruptions.

2. Definition of Cyber Warfare

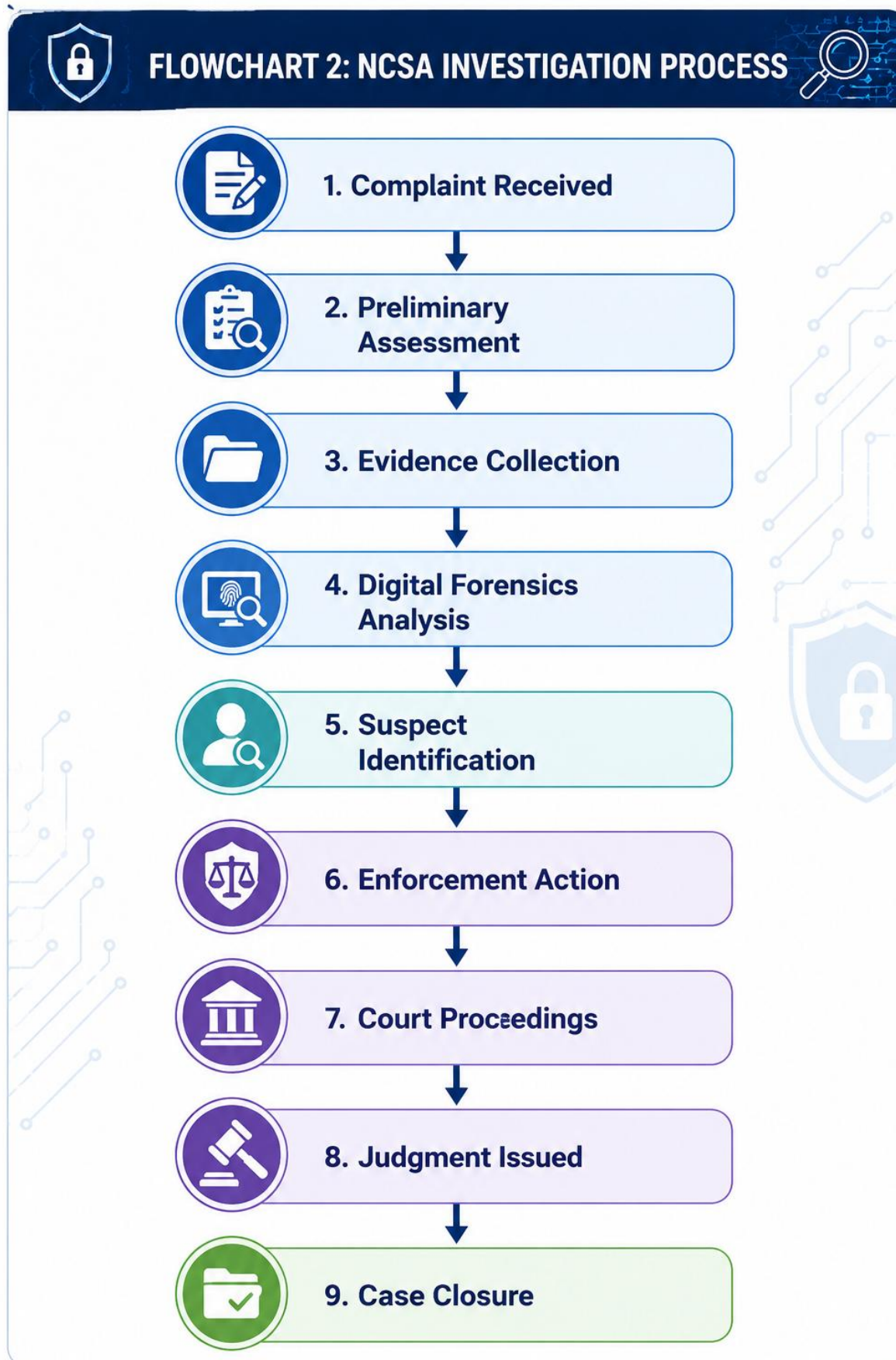
According to the laws of VIT's, cyber warfare will be defined as:

"An act or series of acts carried out or supported by a foreign state, organization, or other entity which causes serious disruption, damage, or threat to national security, public safety, critical infrastructure, or governmental functions."

3. Recommendations

- National Cyber Defense Strategy
A national cyber defense strategy for cyber warfare threats should be developed by the government.
- Classification of National Cyber Incidents
National cyber incidents will be classified as follows:
 - Level 1 - Minor incident.
 - Level 2 - Serious incident.
 - Level 3 - Critical national incident.
 - Level 4 - Cyber war emergency.
- Critical Infrastructure Protection Programs
In addition to these, the following should have additional protection programs:
 - Power grids
 - Health-care systems
 - Transportation infrastructure
 - Government communication system
- National Cyber Reserve Force
The government should form a national cyber reserve force comprising of cybersecurity experts who can help during national cyber emergencies.

Flowchart 2: Investigation Process



Flowchart 3: Data Breach Response



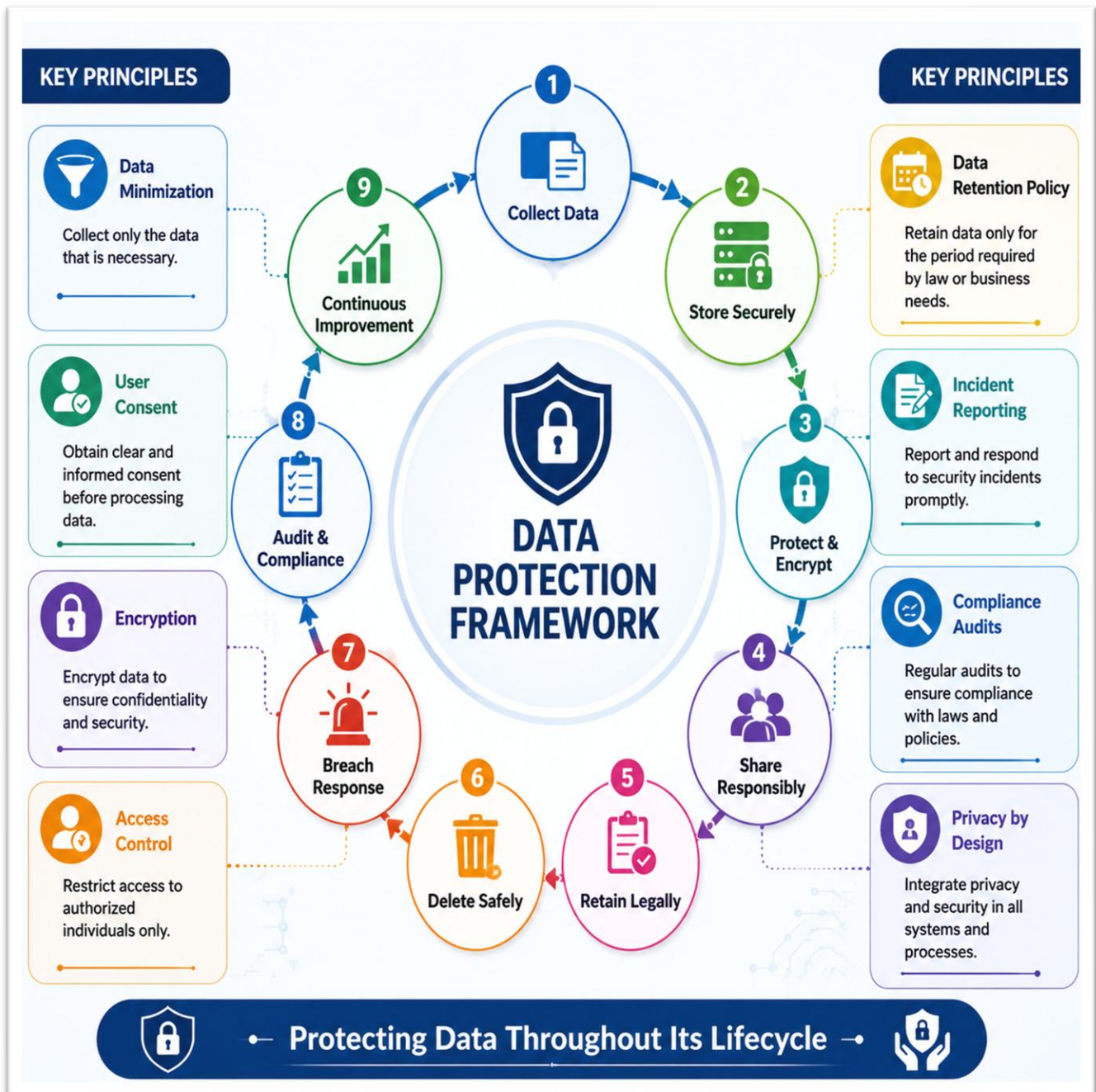
Infographic 1: Citizens' Rights



Infographic 2: Cybercrime Categories



Infographic 3: Security Framework



Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
[Done]	Project Report	Minimum 25 pages, well formatted, all 15 parts completed, diagrams included
[Done]	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
[Done]	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution
[Done]	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment
Done	Deliverable	Requirements
[Done]	Flowchart: Data Breach Response	Shows data breach response procedure for organizations
[Done]	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
[Done]	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
[Done]	Infographic: Security Framework	Visual overview of the data protection and corporate security framework